



Minutes of the 9th meeting of the Board of Studies of the Department of Computer Science, held on 16th May, 2024

The 9th meeting of the board of study of the Department of Computer Science, University of Swabi was held on 16th May, 2024, at 12:00 PM in the Department of Computer Science. The meeting was chaired by Prof. Dr. Mukhtar Alam, Dean Faculty of Science and Dean Faculty of Arts, University of Swabi. The following members attended the meeting:

S.No.	Name of Faculty Member	Designation	University/Institution
1	Prof. Dr. Mukhtar Alam	Dean Faculty of Science	University of Swabi
2	Dr. Shah Nazir	Assistant Professor	University of Swabi
3	Mr. Anwar Hussain	Lecturer	University of Swabi
4	Mr. Yaser Salim	Lecture	University of Swabi
5	Prof. Dr. Sara Shahzad	Professor	University of Peshawar
6	Prof. Dr. Sajid Anwar	Professor	IMSciences, Peshawar

Proceedings of the meeting were started with the recitation of the holy Quran by Dr. Shah Nazir, Assistant Professor, Department of Computer Science. After that, the meeting was formally started by the Dean, Faculty of Science by briefing the members about the agenda items.

The following decisions and recommendations were taken during the meeting along with significant points of discussion.

Agenda 01. Confirmation of minutes of 8th BoS

Background:

The minutes of the 8th meeting of the Board of Study (BoS) were presented to the forum for confirmation.

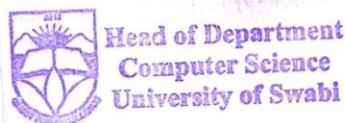
Discussion and Decision:

The forum confirmed the minutes.

Agenda 02. New Course Specification for Bachelor of Studies in Computer Science (BS in Computer Science program)

Background:

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As per the Higher Education Commission (HEC) and National Computing Education and Accreditation Council (NCEAC), the course specifications for courses of BS in Computer Science were presented for discussion and consideration of the forum. The course specifications are required for the purpose of accreditation of the BS in Computer Science program:

Current 3 schemes of studies are being pursued for BS (CS) program. The reasons for following different batches of of BS in Computer Science program. 3 schemes of studies are:

- Schemes of studies are adopted from the HEC and approved by BoS for students enrolled before Fall 2021.
- Scheme of studies adopted for students enrolled in Fall 2021 and Fall 2022 for harmonizing courses launch of AD.
- Scheme of studies adopted for harmonizing courses with the HEC's UG policy 2023

1. Scheme of studies followed for the student that are enrolled before Fall 2021

Semester	Course Code	Course Title	Credit Hours
1 st Semester	CS111	Introduction to Information and Communication Technologies	3(2,1)
	CS112	Programming Fundamentals	4(3,1)
		English Composition and Comprehension	3(3,0)
		Calculus and Analytical Geometry	3(3,0)
		Islamic Studies	2(2,0)
		Total	15(13,2)
2 nd Semester		Linear Algebra	3(3,0)
		Technical and Business Writing	3(3,0)
		Pakistan Studies	2(2,0)
		Applied physics	3(2,1)
	CS121	Object Oriented Programming	4(3,1)
	CS122	Discrete Structures	3(3,0)
		Total	18(16,2)
3 rd Semester	CS211	Data Structure and Algorithms	4(3,1)
		Probability and Statistics	3(3,0)
	CS212	Digital Logic Design	4(3,1)
		University Elective I (Human Resource Management)	3(3,0)
		Communication & presentation skills	3(3,0)
		Total	17(15,2)
4 th Semester	CS221	Comp Organization & Assembly Lang.	4(3,1)
	CS222	Database Systems	4(3,1)
	CS223	Software Engineering	3(3,0)
		University Elective II (Financial Management)	3(3,0)
	CS224	Computer Science Elective I (Web-Design and Development)	3(2,1)
		Total	17(14,3)
5 th Semester	CS311	Operating Systems	4(3,1)
	CS312	Parallel & Distributed Computing	3(3,0)

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	CS313	Theory of Automata	3(3,0)
		Differential Equations	3(3,0)
	CS314	Computer Networks	4(3,1)
		Total	17(15,2)
6 th Semester	CS321	Artificial Intelligence	4(3,1)
	CS322	Compiler Construction	3(3,0)
	CS323	Design and Analysis of Algorithms	3(3,0)
	CS324	Computer Science Elective II (Fundamental of Data Mining)	3(3,0)
	CS325	Computer Science Elective IV (Mobile App Development)	3(2,1)
		Total	16(14,2)
7 th Semester	CS411	Computer Science Elective V (Software Project Management)	3(3,0)
		Professional Practices	3(3,0)
		University Elective III (Entrepreneurship)	3(3,0)
	CS412	Information Security	3(3,0)
	CS413	Final Year Project – I	3(0,3)
		Multivariate Calculus	3(3,0)
		Total	18(15,3)
8 th Semester		Numerical Computing	3(3,0)
	CS421	Human Computer Interaction	3(3,0)
	CS422	Computer Science Elective VI (Natural Language Processing)	3(3,0)
	CS423	Final Year Project – II	3(0,3)
		University Elective IV (Logic and critical Thinking)	3(3, 0)
	CS424	Computer Science Elective III (Object Oriented Analysis and Design)	3(3,0)
		Total	18(15,3)

2. Scheme of studies followed for the student that are enrolled in Fall 2021 and Fall 2022

Semester	Course Code	Subject	Cr. Hrs.
1st Semester	CS111	Introduction to Information and Communication Technologies	3(2-1)
	CS112	Programming Fundamentals	4(3-1)
		English Composition and Comprehension	3(3-0)
		Islamic Studies	3(3-0)
		University Elective I (Civic Education)	3(3-0)
		Total	16(14-2)
2nd Semester		Linear Algebra	3(3-0)
		Technical and Business Writing	3(3-0)
		Pakistan Studies	3(3-0)
		Applied physics	3(2-1)
	CS121	Object Oriented Programming	4(3-1)

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		Total	16(14-2)
3rd Semester	CS211	Data Structure and Algorithms	4(3-1)
		Probability and Statistics	3(3-0)
	CS212	Digital Logic Design	4(3-1)
		University Elective II (Principles of Management)	3(3-0)
		Communication & presentation skills	3(3-0)
		Total	17(15-2)
4th Semester	CS221	Compiler Construction	3(3-0)
	CS222	Design and Analysis of Algorithms	3(3-0)
		University Elective III (Economics)	3(3-0)
	CS223	Web-Design and Development (CSE- I)	3(2-1)
		University Elective IV (Arabic)	3(3-0)
		Total	15(14-1)
5th Semester	CS311	Operating Systems	4(3-1)
		Calculus and Analytical Geometry	3(3-0)
	CS312	Theory of Automata	3(3-0)
		Differential Equations	3(3-0)
	CS313	Computer Networks	4(3-1)
		Total	17(15-2)
6th Semester	CS321	Comp Organization & Assembly Lang.	4(3-1)
	CS322	Database Systems	4(3-1)
	CS323	Artificial Intelligence	4(3-1)
	CS324	Fundamental of Data Mining (CSE- II)	3(3-0)
	CS325	Mobile App Development (CSE- III)	3(2-1)
		Total	18(14-4)
7th Semester	CS411	Software Project Management (CSE- IV)	3(3-0)
		Professional Practices	3(3-0)
	CS412	Parallel & Distributed Computing	3(3-0)
	CS413	Information Security	3(3-0)
	CS414	Final Year Project – I	3(0-3)
		Multivariate Calculus	3(3-0)
8th Semester		Total	18(15,3)
		Numerical Computing	3(3-0)
	CS421	Discrete Structures	3(3-0)
	CS422	Natural Language Processing (CSE- V)	3(3-0)
	CS423	Final Year Project – II	3(0-3)
	CS424	Software Engineering	3(3-0)
		Total	15(12-3)

3. Scheme of studies followed for the student that are enrolled in Fall 2023 and onwards

Semester	Course Code	Subject	Cr. Hrs.
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1st	CS-111	Application of Information & Communication Technologies	3(2-3)
	CS-121	Programming Fundamentals	4(3-3)
	ENG-113	Functional English	3(3-0)
	ISL-111	Islamic Studies	2(2-0)
	POL-162	Civics and Community Engagement	2(2-0)
	CS-101	Mathematics-I (for Pre-Medical students)	
		Applied physics	3(2-3)
		Total	17(14-9)
2nd	-	Linear Algebra	3(3-0)
	ENG-213	Technical and Business Writing	3(3-0)
	POL-122	Ideology and Constitution of Pakistan	2(2-0)
	CS-131	Digital Logic Design	3(2-3)
	CS-122	Object Oriented Programming	4(3-3)
	BBA-202	Entrepreneurship	2(2-0)
	-	Total	17(15-6)
3rd	CS-221	Data Structure	4(3-3)
	STAT-231	Probability and Statistics	3(3-0)
	CS-231	Human Computer Interaction & Computer Graphics	3(2-3)
		Introduction to Management	2(2-0)
	ENG-111	Expository Writing	3(3-0)
	CS-222	Web Technologies (Domain Elective 1)	3(2-3)
	-	Total	18(15-9)
4th	CS-223	Compiler Construction	3(2-3)
	CS-232	Analysis of Algorithms	3(3-0)
	-	Introduction to Marketing (Social Science)	3(3-0)
		Calculus and Analytical Geometry	3(3-0)
	CS-224	Web Engineering (Domain Elective 5)	3(3-0)
	-	Total	15(14-3)
5th	CS-331	Operating Systems	3(2-3)
	CS-321	Software Engineering	3(3-0)
	CS-332	Theory of Automata	3(3-0)
	CS-333	Computer Architecture	3(2-3)
	CS-341	Data Communication and Computer Networks	4(3-3)
	-	Total	16(13-9)
6th	CS-334	Computer Organization & Assembly Language	3(2-3)
	CS-351	Database Systems	4(3-3)
	CS-361	Artificial Intelligence	3(2-3)
	CS-322	Advance Programming – Visual Pro	3(2-3)
	CS-323	Mobile App Development	3(2-3)
	-	Total	16(11-15)
7th	CS-421	Software Testing & Quality Assurance (Domain Elective)	3(2-3)
	-	Professional Practices	2 (2-0)
	CS-431	Parallel & Distributed Computing	3(2-3)
	CS-441	Information Security	3(2-3)
	-	Multivariable Calculus	3(3-0)
	CS-598	Final Year Project – I	2(0-6)

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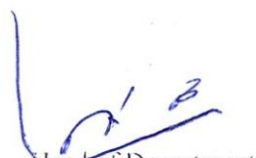
	-	Total	16(11,15)
8th	MATH-331	Numerical Analysis	3(2-3)
	CS-432	Discrete Structures	3(3-0)
	CS-442	Cyber Security	3(2-3)
	CS-452	Advance database management system	3 (2-3)
	CS-599	Final Year Project – II	4(0-12)
	-	Total	16(9-21)

Discussion and Decision:

Mr. Yaser Salim, the focal person for NCEAC accreditation of BS program has shared the detail of progress and updates regarding accreditation as per the updated information required in the NCEAC online portal. The Programme Learning Outcomes (PLOs), Course Learning Outcomes (CLOs), and GA (Graduate Attributes) were discussed. The members of the forum have suggested the following recommendations:

- The external member of the BoS suggested to wait for initiating accreditation as the various meetings are in progress under the HEC and HED to defined the universally accepted CLOs and PLOs to be adopted by universities. In case of their final recommendation, the department has to inline their CLOs and PLOs again that will result in repeating the same efforts for such activity.
- One the other hand, for safe side, the department was in favor of initiating the process for NCEAC visit as the NCEAC has already given deadlines to the universities. So, the forum agreed for initiating NCEAC accreditation process for BS program and suggested to include the proposed CloS, PloS, and GA by the department in the online system.
- The department is following HEC curriculum 2017, and 2023 as the policy has revised various time for incorporating AD in BS programme and adaptation of undergraduate policy 2023. Keeping in view the above requirement, the forum suggested that for harmonizing their CLOs and PLOs, the first priority should be given to the latest adopted HEC 2023 undergraduate education policy and if there is any shortfall of course(s) then the undergraduate policy 2017 be adopted. The proposed course specification for Bachelor of Studies in Computer Science is Annexed (Annex-A).

The meeting concluded with vote of thanks by the convener.


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